

Player--gameweek panel

- Player, club, and position
- Price, minutes, and points
- Historical performance features

Preprocessing

- GW1--26 training window
- Type checks and deduplication
- Leakage-safe feature construction

Exploratory analysis

- Position-specific correlations
- Feature redundancy
- Distributions and SHAP values

Forecasting

- Averages and smoothing
- ARIMA and simulation
- Regularized feature models

Optimization

- Two-stage baseline
- Joint 15-player MILP
- Pareto and robust variants

Outputs and evaluation

- Squad, XI, captain, and bench
- Fixed-squad lineup updating
- GW27--38 scores and diagnostics

